Avijnan Purkait

+91 62909 86642 | avijnanpurkait@gmail.com | linkedin.com/in/avijnan-purkait-6ba580318 | github.com/avi-dev-machine
Portfolio: avijnanpurkait.vercel.app | Kolkata, West Bengal

SKILLS

Languages & Tools: Python, C, Java, SQL, Git, GitHub, Docker
AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Small Language Models (SLMs), Fine-tuning, Quantization
Agentic AI & LLMs: LangGraph, Multi-Agent Systems, Tool Calling, Hybrid RAG, GraphRAG, Knowledge Graphs, Prompt Engineering, LLM APIs
Libraries: TensorFlow, TensorFlow Lite, PyTorch, ONNX Runtime, Scikit-learn, OpenCV, YOLO, NumPy, Pandas
Backend & Edge: FastAPI, Flask, REST APIs, SQLite, Raspberry Pi, ESP32, LoRa, IoT Sensors

EDUCATION

Heritage Institute of Technology, Kolkata, West Bengal 2023 – Present
B.Tech in Computer Science & Engineering (AI & ML) — GPA: 7.75/10
Dolna Day School, Kolkata — Class XII (ISC): 85.5%; Class X (ICSE): 94.4% 2023 / 2021

EXPERIENCE

Technical Project Lead — Agentic AI for Industrial Intelligence, BMC Electroplast 2026 – Present
Industry–Academia Collaboration | LangGraph, Multi-Agent Systems, Hybrid RAG, GraphRAG, Knowledge Graphs

- Leading a 10-member team of third-year CSE (AI & ML) students building an agentic AI platform for industrial intelligence; owning multi-agent design and task decomposition.
- Designing a supervisor–worker architecture with specialized agents for production analysis, quality inspection, predictive maintenance, document intelligence and root-cause investigation, with intent classification, planning and conditional routing.
- Building tool-augmented agents combining data analysis, hybrid RAG/GraphRAG (multi-hop knowledge-graph reasoning) and computer-vision outputs for evidence-grounded, source-attributed answers.
- Designing stateful workflows with checkpointing, bounded loops, failure recovery, tool-result validation and human-in-the-loop approval, plus evaluation and observability for retrieval relevance, tool-call accuracy and hallucination detection.

Software Development Intern — BDS Cube Technologies Jun 2026 – Aug 2026

- Building a client-facing DPDP Act, 2023 compliance platform: FastAPI, RAG-based search, structured JSON datasets and REST APIs connecting legal documents, semantic retrieval, AI-assisted guidance and the web frontend.

PROJECTS

SolveX — AI Reasoning & Visualization Workspace | LangGraph, FastAPI, Next.js, p5.js | Live Demo *Final Year Project*

- Built a dual-agent technical reasoning workspace: a LangGraph agent routes queries to Maths, ML, Automata and Algorithms engines and streams stepwise LaTeX solutions with a typed viz_state; a second agent turns that state into deterministic Mermaid.js diagrams and p5.js animations.
- Unlike general-purpose chat LLMs, whose diagrams can contradict their own text, visuals derive from one typed state and pass a consistency gate (static fallback), Playwright validation and a sandbox for generated Python.
- Designed a constrained micro-DSL runtime so lightweight free-tier LLMs (Gemini Flash, Llama-3.3 on Groq) can produce animations; added SSE streaming and text, voice and PDF-crop input.

Agricultural Intelligence System | Raspberry Pi, SLMs, RAG, YOLO, OpenCV, SQLite, LoRa

- Deployed quantized small language models on Raspberry Pi for offline crop advisory, symptom interpretation and report summarization, using prompt compression, context trimming and local caching for CPU-only inference.
- Curated a cleaned, deduplicated agriculture instruction dataset and generated synthetic data from extension guidelines and crop manuals.
- Built a RAG layer grounding SLM output in local crop guidelines and historical observations, with structured output schemas (risk level, actions, evidence) to reduce hallucination.
- Fused SLM reasoning with crop-health vision, IoT, weather and satellite data on an offline-first SQLite architecture.

FitVision | FastAPI, YOLO11n-Pose, OpenCV, Next.js, Gemini, Groq *Smart India Hackathon Finalist*

- Built a real-time pose estimation pipeline using YOLO11n-Pose and OpenCV, tracking body keypoints to count reps and evaluate form across 8 exercises at 15+ FPS on CPU.
- Designed a pixel-to-centimetre calibration algorithm estimating jump height and reach with approx. 92% accuracy vs. manual measurements.
- Integrated Gemini and Groq APIs for AI performance reports and nutrition recommendations in <3 s; deployed with Docker on Hugging Face Spaces, supporting 10+ concurrent WebSocket users at <200 ms latency; SIH 2025 Finalist.

SETUKA | ESP32, LoRa Mesh, FastAPI, GenAI, Isolation Forest *Hult Prize Finalist, IIT Bombay*

- Engineered a LoRa Mesh network across ESP32 nodes for emergency communication in no-cellular zones, achieving 3 km/hop with approx. 95% delivery reliability.
- Developed an Isolation Forest anomaly-detection model (approx. 88% precision) and a FastAPI backend with GenAI incident reports and real-time mesh coordination.
- Contributed to product strategy, leading to a Hult Prize India 2026 Finalist finish and a provisional patent.

CERTIFICATIONS

NPTEL: Programming with Generative AI (Ongoing); Ethical Hacking (Ongoing) 2026

LEADERSHIP AND COLLABORATION

- **Team Lead** – Led cross-functional teams (ML, embedded systems, business) on FitVision, SETUKA and the BMC agentic AI platform.
- **Coordinator** – Geeks United, college coding club spanning hardware and software wings.
- **Organizing Committee Member** – Logistics, judging panels and technical execution for college hackathons.
- **Fest Council Member** – Officially elected to the Fest Council of the college fest body twice (technical and cultural fests).
- **Captain** – Heritage Institute of Technology Football Team; directed on-field strategy and mentored teammates.

ACHIEVEMENTS

- Finalist – Hult Prize India 2026 (IIT Bombay); Smart India Hackathon 2025; IIT Kharagpur Data Science Hackathon 2025
- Winner – Hult Prize, HIT 2026; 2nd Runner-Up – IIC Innovation Challenge 2026, HIT